

A Branch-Regular Tree Reduction for Large Almost Disjoint Families on $\mathbb{N}^*$

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Status

Conditional reduction, likely valid. The proof below is complete assuming the explicit tree principle in Definition 1. The packet does not prove that this tree exists in ZFC.

Source Question

Lambie-Hanson–Schrittesser [1] define an almost disjoint family on a topological space X to be a family of noncompact regular open subsets of X whose pairwise intersections are compact. Question 4.3 asks whether ZFC proves the existence of such a family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even $2^{2^{\aleph_0}}$.

operator $E : c_0(\omega_1) \rightarrow \ell_\infty([\omega]^\omega)$: in [2, Theorem 1], it is proven that CH implies that $\text{Ext}^2(c_0(\omega_1), c_0) \neq 0$. It is at least conceivable that the opposite answer might follow from an assumption such as the Proper Forcing Axiom.

Finally, we wonder whether an assumption like $\mathfrak{b} = 2^{\aleph_0}$ is necessary to construct a large almost disjoint family on $\mathbb{N}^*$.

Question 4.3. *Is it provable in ZFC that there exists an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even of cardinality $2^{2^{\aleph_0}}$?*

Conditional Dependencies

The only unproved dependency is the following tree principle.

Definition 1 (Binary branch-regular Hausdorff tree). *A binary branch-regular Hausdorff tree in $\mathcal{P}(\omega)/\text{Fin}$ is a family*

$$\langle a_s : s \in 2^{<\omega_1} \rangle$$

of infinite subsets of ω satisfying:

1. *If $s \subsetneq t$, then $a_s \subseteq^* a_t$ and $a_t \setminus a_s$ is infinite.*
2. *If $s, t \in 2^{<\omega_1}$ are distinct, then*

$$a_s \cap a_t = {}^*a_{s \wedge t},$$

where $s \wedge t$ is their longest common initial segment.

3. *For every $s \in 2^{<\omega_1}$ and $i \in \{0, 1\}$, $a_{s \smallfrown i} \setminus a_s$ is infinite.*

4. For every branch $f \in 2^{\omega_1}$ and every infinite $y \subseteq \omega$, either

$$y \subseteq^* a_{f \upharpoonright \alpha}$$

for some $\alpha < \omega_1$, or there is an infinite $z \subseteq y$ such that

$$z \cap a_{f \upharpoonright \alpha} = {}^*\emptyset \quad \text{for all } \alpha < \omega_1.$$

The last clause is the branch-regularity clause. It is exactly the place where the source paper's $\mathfrak{b} = 2^{\aleph_0}$ hypothesis performs work in its construction of a stronger ω -branching tree.

Idea of the Proof

Given a branch f , take the open subset of $\mathbb{N}^*$ generated by the clopens $N_{a_{f \upharpoonright \alpha}}$. Branch-regularity makes this open set regular open. Strict growth along the branch makes it noncompact. If two branches split at a node r , the controlled-intersection clause forces the intersection of the two corresponding open sets to be precisely the clopen N_{a_r} . Thus the branches of the tree become the desired generalized almost disjoint family.

Theorem 1. *If there exists a binary branch-regular Hausdorff tree in $\mathcal{P}(\omega)/\text{Fin}$, then there is an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$.*

Proof. For $a \in [\omega]^\omega$, let

$$N_a = \{p \in \mathbb{N}^* : a \in p\}$$

be the corresponding basic clopen subset of $\mathbb{N}^*$. Let $\langle a_s : s \in 2^{<\omega_1} \rangle$ be a binary branch-regular Hausdorff tree. For each branch $f \in 2^{\omega_1}$, define

$$U_f = \bigcup_{\alpha < \omega_1} N_{a_{f \upharpoonright \alpha}}.$$

This set is open.

We first show that U_f is regular open. Let $p \in \mathbb{N}^* \setminus U_f$, and let N_y be a basic clopen neighborhood of p , with $y \in [\omega]^\omega$. Since $p \notin U_f$, we cannot have $y \subseteq^* a_{f \upharpoonright \alpha}$ for any $\alpha < \omega_1$; otherwise $a_{f \upharpoonright \alpha} \in p$, so $p \in U_f$. By branch-regularity, choose an infinite $z \subseteq y$ such that

$$z \cap a_{f \upharpoonright \alpha} = {}^*\emptyset \quad \text{for all } \alpha < \omega_1.$$

Then $N_z \subseteq N_y$ and $N_z \cap U_f = \emptyset$. Thus every basic neighborhood of every point outside U_f contains a smaller basic clopen set disjoint from U_f . Equivalently, no point of $\mathbb{N}^* \setminus U_f$ lies in $\text{int cl}(U_f)$. Since U_f is open, it follows that $U_f = \text{int cl}(U_f)$, so U_f is regular open.

Next U_f is not compact. The family

$$\{N_{a_{f \upharpoonright \alpha}} : \alpha < \omega_1\}$$

is an increasing open cover of U_f . No finite subfamily covers U_f : if $\alpha < \omega_1$, then $a_{f \upharpoonright (\alpha+1)} \setminus a_{f \upharpoonright \alpha}$ is infinite, so there is a nonprincipal ultrafilter containing this difference; it lies in $N_{a_{f \upharpoonright (\alpha+1)}} \setminus N_{a_{f \upharpoonright \alpha}}$. Hence U_f is noncompact.

Now let $f \neq g$ be two branches. Let $\delta < \omega_1$ be the first ordinal with $f(\delta) \neq g(\delta)$, and put $r = f \upharpoonright \delta = g \upharpoonright \delta$. We claim that

$$U_f \cap U_g = N_{a_r}.$$

The inclusion $N_{a_r} \subseteq U_f \cap U_g$ follows from $a_r \subseteq^* a_{f \upharpoonright \alpha}$ and $a_r \subseteq^* a_{g \upharpoonright \beta}$ for all sufficiently large $\alpha, \beta > \delta$.

Conversely, if $p \in U_f \cap U_g$, choose $\alpha, \beta < \omega_1$ such that $a_{f \upharpoonright \alpha} \in p$ and $a_{g \upharpoonright \beta} \in p$. Then

$$a_{f \upharpoonright \alpha} \cap a_{g \upharpoonright \beta} \in p.$$

By the controlled-intersection clause, this intersection is equal modulo finite to $a_{(f \upharpoonright \alpha) \wedge (g \upharpoonright \beta)}$, and the latter is equal modulo finite to a_r after increasing α, β past $\delta + 1$. Since p is nonprincipal, $a_r \in p$, so $p \in N_{a_r}$. Thus $U_f \cap U_g = N_{a_r}$, a clopen and therefore compact subset of $\mathbb{N}^*$.

Finally, the sets U_f are distinct for distinct branches. If f, g split at r and $s = r \frown f(\delta)$, then $a_s \setminus a_r$ is infinite and is almost disjoint from every node on the branch g above r . A nonprincipal ultrafilter containing $a_s \setminus a_r$ belongs to $U_f \setminus U_g$. Hence $\{U_f : f \in 2^{\omega_1}\}$ has cardinality $2^{\aleph_1}$.

We have produced $2^{\aleph_1}$ many noncompact regular open subsets of $\mathbb{N}^*$ with compact pairwise intersections, as required. $\square$

Verification and Remaining Work

The proof above uses the tree dependency only in three places: branch regularity gives regular openness, strict branch growth gives noncompactness, and controlled intersections give compact pairwise intersections. Without the dependency, no conclusion is claimed.

To turn this conditional packet into a full solution of the first part of Question 4.3, it remains to construct the binary branch-regular Hausdorff tree in ZFC. The source paper constructs a related stronger tree under $\mathfrak{b} = 2^{\aleph_0}$; removing that hypothesis is the open core.

Literature and Search Notes

The local run indexes were checked for arXiv:2606.02040 and the existing continuum-sized partial packet. Web searches were run for Hausdorff gaps, regular open subsets of ω^* , almost disjoint families on ω_1 , and branch/tree formulations in $\mathcal{P}(\omega)/\text{Fin}$. The searches did not reveal a ZFC construction of the dependency above. They did reinforce that Hausdorff and Luzin gaps are the natural nearby tools.

Human Review Recommendation

Reviewers should focus on the conditional dependency. The implication from the tree to the topological family is straightforward; the real mathematical question is whether the branch-regular tree can be built in ZFC, or whether its existence implies a cardinal-invariant hypothesis such as a domination principle.

References

- [1] C. Lambie-Hanson and D. Schritterser, *Generalized almost disjoint families and injective Banach spaces*, arXiv:2606.02040, 2026. <https://arxiv.org/abs/2606.02040>